

Khawaja Yunus Ali University

School of Science and Engineering

Department of Computer Science and Engineering

Final Examination: Spring Semester 2019

Program: B.Sc. in CSE (Batch: 7th) 1st Year 2nd Semester

Course Title: Physics – II

Course Code: PHY 1201

Total Time: 2 hours

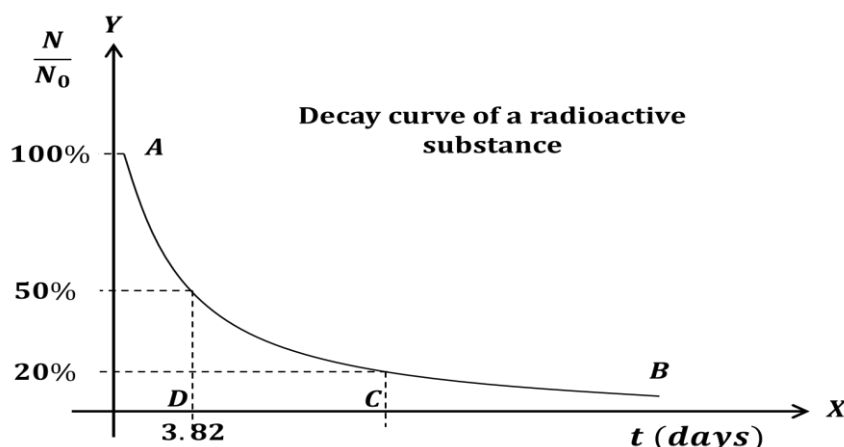
Full Marks: 40

Part-B

[Answer the question No. 1 and any four (04) from the rest]

5 × 7 = 35

1. See the stem given below and answer the following questions:



- | | |
|---|---|
| a) Define the point D. | 1 |
| b) Find the value of point C. | 4 |
| c) Is it possible to touch the point B with X axis? Justify your answer. | 2 |
| 2. a) What is Alpha particle? 1 | |
| b) Prove that electron cannot exist inside the nucleus. | 3 |
| c) What will happen, when a particle moves with the speed of light? | 3 |
| 3. a) "For the polarizing angle of incidence, the reflected & refracted rays are 90° apart" – explain the statement in according with Brewster law. 4 | |
| b) The refractive index for a plastic is 1.25. calculate the angle of refraction for a ray of light incident at polarizing angle. | 3 |
| 4. a) Define Wave front. 1 | |
| b) What is diffraction grating? Find the grating constant of diffraction grating having 1000 lines per cm . | 3 |
| c) Light from a lamp ($\lambda = 5890nm$) forms an interference pattern on a screen $0.8m$ from a pair of slits. The bright fringes in the pattern are $0.35cm$ apart. What is the slit separation? | 3 |

P.T.O

5. a) What is threshold frequency? 1
- b) Write down the condition for occurring photoelectric effect in metal. 2
- c) UV light of wavelength 350 nm and intensity 1.00 W/m^2 is directed at a potassium surface. Find the maximum K.E. of photoelectrons. (Work function of Potassium is 2.2 eV) 4
6. a) Draw the binding energy per nucleon curve. 1
- b) Calculate the radius and energy of the third orbit of hydrogen atom. 3
- c) Fission reaction - ${}_0^1n + {}_{92}^{235}\text{U} \rightarrow [{}_{92}^{236}\text{U}] \rightarrow {}_{56}^{141}\text{Ba} + {}_{36}^{92}\text{Kr} + 3{}_0^1n + E$
- Where, $M({}_{92}^{235}\text{U}) = 235.0439 \text{ amu}$
- $M({}_{56}^{141}\text{Ba}) = 140.9139 \text{ amu}$
- $M({}_{36}^{92}\text{Kr}) = 91.8973 \text{ amu}$
- $M({}_0^1n) = 1.0087 \text{ amu}$
- Find the value of E . 3

P.T.O.

8. The angular momentum of an electron in the orbit of an atom is –

a) $mvr = \frac{h}{2}$

b) $mvr = \frac{n}{2}$

c) $mvr = \frac{\pi}{2}$

d) $mvr = \frac{nh}{2\pi}$

e) $mvr = \frac{nh}{2}$

9. What is the speed of gamma ray?

a) $3 \times 10^8 \text{ ms}^{-1}$

b) $3 \times 10^{-8} \text{ ms}^{-1}$

c) 332 ms^{-1}

d) Zero

e) Infinity

10. The energy of an electron in the 1st orbit of the hydrogen atom is

a) 36eV

b) 17.6eV

c) 1.12eV

d) 0.53eV

e) -13.6eV

Invigilator Signature

Examiner Signature